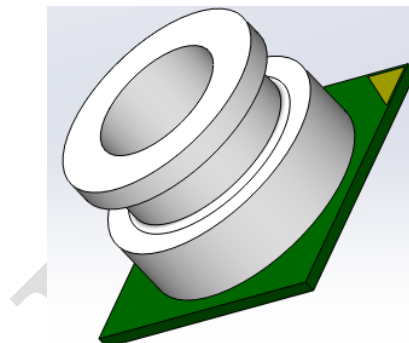


PRECISION BAROMETER AND ALTIMETER SENSOR

Key Features

- Operation temperature: -40~85℃
- Supply voltage: 1.8V to 3.6V
- Pressure range: 0 ~1000Kpa
- Programmable events and interrupt controls
- High-speed I2C digital output interface (Up to 10 MHz)
- Fast conversion down to 2ms
- Standby current <0.1uA, low conversion power, 2uA at 1 sample/sec.
- Water proof design can use in 100m



4.5x4.5x3.0mm

Applications

- Depth meter systems
- Industrial Pressure and Temperature Sensor System
- Adventure and Sports watches
- Diving computers
- Data loggers for pressure, temperature
- Panel computer
- Hermetically sealable for outdoor devices

Descriptions

The HP5834-10BA employs a MEMS pressure sensor with an I2C interface to provide accurate Temperature, Pressure data. The sensor Pressure and Temperature outputs are digitized by a high resolution 24-bit ADC. Data compensation is integrated internally to save the effort of the external host MCU system. Easy command-based data acquisition interface and programmable interrupt control is available. Typical active supply current is 5.3uA per measurement-second while the ADC output is filtered and decimated by 256.

1. Typical Application Circuit

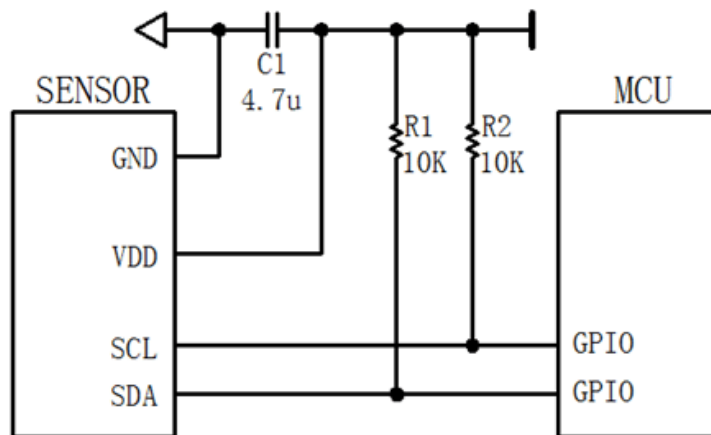
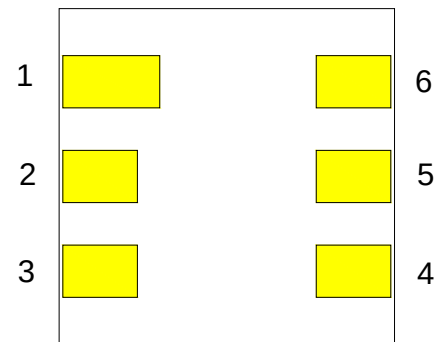


Figure 1 Typical Application Circuit

2. Pin Configuration

Table1 Pin Descriptions

Pin	Name	I/O	Function
1	GND	I	Ground
2	INT1	O	Interrupt 1 output pin
3	SDA	IO	I ² C serial bi-directional data pin
4	SCL	IO	I ² C serial clock input pin
5	NC	-	NO Connect
6	VDD	I	Power supply



BOTTOM VIEW

3. Mechanical and Electrical Specifications

3.1 Pressure and Temperature Characteristics

Table2 Pressure Output Characteristics @ VDD = 3.0V, T = 25°C unless otherwise noted

Parameter	Symbol	Conditions	Min	Typ.	Max	Unit
Pressure Measurement Range	P _{FS}	Absolute	0		1000	Kpa
Pressure Absolute Accuracy		0 to 1000 Kpa from 0°C to 40°C	-2.5		+2.5	Kpa
		0 to 1000 Kpa from -20°C to 60°C	-5.0		+5.0	Kpa
Pressure Relative Accuracy		0 to 1000 Kpa at 25°C		±1.0		Kpa
		0 to 1000 Kpa From -20°C to 60°C		±2.0		
Max Error with Power Supply		Power supply from 1.8V to 3.6V	-1.0		+1.0	Kpa
Pressure Resolution		Pressure Mode		0.01		Kpa
Board Mount Drift		After solder reflow		±1.5		Kpa
Long Term Drift		After a period of 1 year		2.0		Kpa

Table3 Temperature Output Characteristics @ VDD = 3.0V, T = 25°C unless otherwise noted

Parameter	Symbol	Conditions	Min	Typ	Max	Unit
Operation Temperature	T _{OP}		-40		85	°C
Temperature Absolute Accuracy		25°C		±1.0		°C
		-10°C to +70°C		±3.0		°C
Max Error with Power Supply		Power supply from 1.8V to 3.6V		±0.5		°C
Temperature Resolution of Output Data				0.01		°C

3.2 Electrical Characteristics

Table4 DC Characteristics @VDD=3.0 V, T=25°C unless otherwise note

Parameter	Symbol	Conditions	Min	Typ	Max	Unit
Operation Supply Voltage	V _{DD}		1.8	3.0	3.6	V
Operation Temperature	T _{OP}		-40		85	°C
Average Operation Current (Pressure Measurement under One Conversion per Second)	I _{DDAVP}	OSR*		85.2		μA
				42.6		
				21.3		
				10.7		
				5.3		
				2.7		
Average Operation Current (Temperature Measurement under One Conversion per Second)	I _{DDAVT}	OSR*		68.8		μA
				34.4		
				17.2		
				8.6		
				4.3		
				2.2		
Conversion Time of Pressure or Temperature	t _{CONV}	OSR*		65.6		ms
				32.8		
				16.4		
				8.2		
				4.1		
				2.1		
Peak Current	I _{PEAK}	During conversion		1.3		mA
Standby Supply Current	I _{DDSTB}	At 25°C			0.1	μA
Serial Data Clock Frequency	f _{SCLK}			100	400	kHz
Digital Input High Voltage	V _{IH}		0.8			V
Digital Input Low Voltage	V _{IL}				0.2	V
Digital Output High Voltage	V _{OH}	IO = 0.5 mA	0.9			V
Digital Output Low Voltage	V _{OL}	IO = 0.5 mA			0.1	V
Input Capacitance	C _{IN}			4.7		uF

*OSR stands for over sampling rate

3.3 Absolute Maximum Rating

Table 5 Maximum Rating

Parameter	Symbol	Conditions	Min	Max	Unit
Overpressure	P_{MAX}			2	Mpa
Supply Voltage	V_{DD}		-0.3	3.6	V
Interface Voltage	V_{IF}		-0.3	$V_{DD}+0.3$	V
Storage Temperature Range	T_{STG}		-40	150	°C
Maximum Soldering Temperature	T_{MS}	40 second maximum		250	°C
ESD Rating		Human body model	-2	+2	kV
Latch-up Current		At 85°C	-100	100	mA

Stresses above those listed as “absolute maximum ratings” may cause permanent damage to the device. This is a stress rating only and functional operation of the device under these conditions is not implied. Exposure to maximum rating conditions for extended periods may affect device reliability.

4. Function Descriptions

4.1 General Description

The HP5834 is a high precision barometer, the pressure and the temperature by an internal 24-bit ADC and compensates them by a patented algorithm. The fully-compensated values can be read out via the I²C interface by external MCU.

4.2 Factory Calibration

Every device is individually factory calibrated for sensitivity and offset for both of the temperature and pressure measurements. The trim values are stored in the on-chip 128-Byte Non-Volatile Memory (NVM). In normal situation, further calibrations are not necessary to be done by the user.

4.3 Automatic power on initialization

Once the device detects a valid VDD is externally supplied, an internal Power-On-Reset (POR) is generated and the device will automatically enter the power-up initialization sequence. After that the device will enter the sleep state. Normally the entire power-up sequence consumes about 400 us.

The user can scan a DEV_RDY bit in the INT_SRC register in order to know whether the device has finished its power-up sequence. This bit appears to 1 when the sequence is done. The device stays in the sleep state unless it receives a proper command from the external MCU. This will help to achieve minimum power consumptions.

4.4 Sensor Output Conversion

For each pressure measurement, the temperature is always being measured prior to pressure measurement automatically, while the temperature measurement can be done individually. The conversion results are stored into the embedded memories that retain their contents when the device is in the sleep state.

The conversion time depends on the value of the OSR parameter sent to the device within the ADC_CVT command. Six options of the OSR can be chosen, range from 128, 256 ... to 4096. The below table shows the conversion time according to the different values of OSR:

Table 6 Conversion Time VS OSR

OSR	Conversion Time (ms)	
	Temperature	Temperature and Pressure
128	2.1	4.1
256	4.1	8.2
512	8.2	16.4
1024	16.4	32.8
2048	32.8	65.6
4096	65.6	131.1

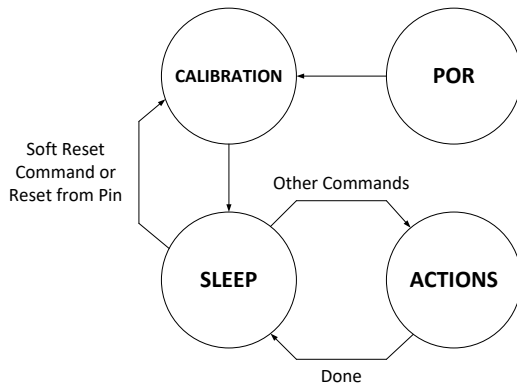
The higher OSR will normally achieve higher measuring precision, but consume more time and power.

The conversion results can be compensated or uncompensated. The user can enable/disable the compensation by setting the ADC_CVT register before performing the conversions.

5. Access Modes & Commands

5.1 Operation Flow

During each power-up/reset cycle, the device will only perform one calibration. After that it will enter the SLEEP state waiting for any incoming commands. It will take actions after receiving different proper commands, and re-enters the SLEEP state when it finishes the jobs.



5.2 Command

The Command Set (Table 6) allows the user to control the device to perform the measuring, results reading and the miscellaneous normal operations.

Table7 Command set

Name	Hex Code	Binary Code	Descriptions
SOFT_RST	0x06	0000 0110	Soft reset the device
ADC_CVT	NA	010_OSR_chnl	Perform ADC conversion
READ_PT	0x10	0001 0000	Read the temperature and pressure values
READ_P	0x30	0011 0000	Read the pressure value only
READ_T	0x32	0011 0010	Read the temperature value only
ANA_CAL	0x28	0010 1000	Re-calibrate the internal analog blocks

5.2.1 Soft Reset the Device

SOFT_RST (0x06)

Once the user issues this command, the device will immediately be reset no matter what it is working on. Once the command is received and executed, all the memories (except the NVM) will be reset to their default values following by a complete power-up sequence to be automatically performed.

5.2.2 OSR and Channel Setting

ADC_CVT (010, 3-bit OSR, 2-bit CHNL)

This command let the device to convert the sensor output to the digital values with or without compensation depends on the PARA register setting. The 2-bit channel (CHNL) parameter tells the device the data from which channel(s) shall be converted by the internal ADC. The options are shown below:

- 00: sensor pressure and temperature channel
- 10: temperature channel

The 3-bit OSR defines the decimation rate of the internal digital filter as shown below:

000:	OSR = 4096	011:	OSR = 512
001:	OSR = 2048	100:	OSR = 256
010:	OSR = 1024	101:	OSR = 128

Setting the CHNL bits to the value of 01 or 11, or the OSR bits to the values of 110 or 111 will lead to failure of conversion.

5.2.3 Read the Temperature and Pressure Values

READ_PT (0x10)

The temperature data is arranged as 20-bit 2's complement format and the unit is in degrees C. Temperature value is stored in all 24 bits of OUT_T_MSB, OUT_T_CSB and OUT_T_LSB. The 4 most significant bits of the 24-bit data is useless, while the 20 least significant bits represent the temperature value. The user shall convert this 20-bit 2's complement binary value into an integer, and then divide the integer by 100 to obtain the final result.

The pressure data is arranged as 20-bit the unit is in Pascal. Pressure value is stored in all 24 bits of OUT_T_MSB, OUT_T_CSB and OUT_T_LSB. The 4 most significant bits of the 24-bit data is useless, while the 20 least significant bits represent the pressure value. The user shall convert this 20-bit unsigned binary value into an integer.

For Example: (Temperature)

Hex value	OUT_T_MSB	OUT_T_CSB	OUT_T_LSB	Dec value
0x000A5C	0x00	0x0A	0x5C	26.52
0xFFFC02	0xFF	0xFC	0x02	-10.22

For Example: (Unsigned data pressure)

P(Units: Kpa)

Hex value	OUT_P_MSB	OUT_P_CSB	OUT_P_LSB	Dec value
0x00A880	0x00	0xA8	0x80	43136
43136 / 100 = 431.36 Kpa				

5.2.4 Read the Pressure Value

.READ_P (0x30)

The pressure data is arranged as 20-bit the unit is in Pascal. Pressure value is stored in all 24 bits of OUT_T_MSB, OUT_T_CSB and OUT_T_LSB. The 4 most significant bits of the 24-bit data is useless, while the 20 least significant bits represent the pressure value. The user shall convert this 20-bit unsigned binary value into an integer.

5.2.5 Read the Temperature Value

.READ_T (0x32)

The temperature data is arranged as 20-bit 2's complement format and the unit is in degrees C. Temperature value is stored in all 24 bits of OUT_T_MSB, OUT_T_CSB and OUT_T_LSB. The 4 most significant bits of the 24-bit data is useless, while the 20 least significant bits represent the temperature value. The user shall convert this 20-bit 2's complement binary value into an integer, and then divide the integer by 100 to obtain the final result.

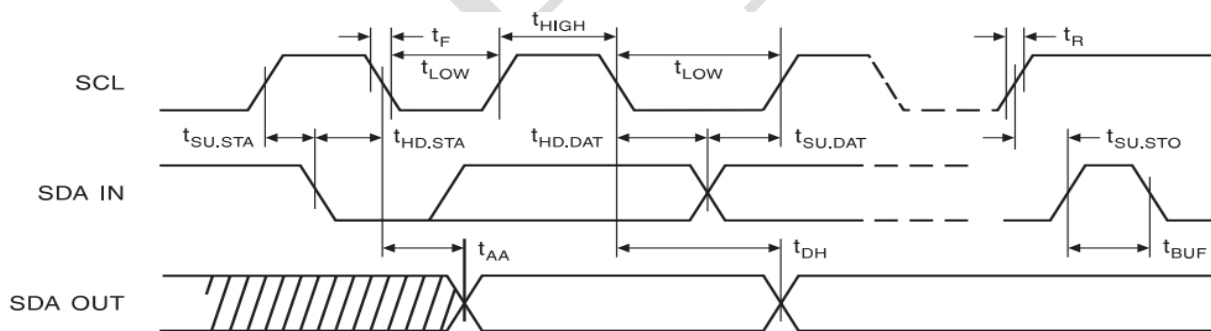
6. I²C Interface

The I²C interface is fully compatible to the official I²C protocol specification. All the data are sent starting from the MSB. Successful communication between the host and the device via the I²C bus can be done using the four types of protocol introduced below.

6.1 I²C Specification

Table8 I²C Slave Timing Values

Parameter	Symbol	I ² C				Unit
		Condition	Min	Typ.	Max	
SCL Clock Frequency	SCL	Pull-up = 10 kΩ	0		400	KHz
Bus free time between STOP and START condition	t _{BUF}		1.5			μs
Repeated START Hold Time	t _{HD.STA}		0.6			μs
Repeated START Setup Time	t _{SU.STA}		0.6			μs
STOP Condition Setup Time	t _{SU.STO}		0.6			μs
SDA Data Hold Time	t _{HD.DAT}		100			ns
SDA Setup Time	t _{SU.DAT}		100			ns
SCL Clock Low Time	t _{LOW}		1.5			μs
SCL Clock High Time	t _{HIGH}		0.6			μs
SDA and SCL Rise Time	t _R		30		500	ns
SDA and SCL Fall Time	t _F		30		500	ns



6.2 I²C Device and Register Address

The I²C device address is shown below. The LSB of the device address is corresponding to address 0XEC (write) and 0XED (read).

A7	A6	A5	A4	A3	A2	A1	W/R
1	1	1	0	1	1	0	0/1

6.3 I²C Protocol

The 1st TYPE: the host issuing a single byte command to the device

The host shall issue the Device Address (ID) followed by a Write Bit before sending a Command byte. The device will reply an ACK after it received a correct SOFT_RST command.

	1	1	1	0	1	1	0	0	0	0	0	0	0	0	1	1	0	0	
S	Device Address							W	A	Command								A	P

The 2nd TYPE: the host writing a register inside the device

The host shall issue the Device Address (ID) followed by a Write Bit before sending a command byte and a data byte. This format only applies while the user wants to send the WRITE_REG command.

	1	1	1	0	1	1	0	0	0	1	1	0	0	1	0	1	0	0	0	0	0	1	1	0	0	
S	Device Address							W	A	Command								A	Data						A	P

The 3rd TYPE: the host reading a register from the device

In this activity there are two frames that are sent separately. The first frame is to send the READ_REG command which contains a 2-bit binary number of 10 followed by a 6-bit register address. The format of the first frame is identical to the 1st type activity. In the second frame, the device will send back the register data after receiving the correct device address followed by a read bit. This format only applies while the user wants to use the READ_REG command.

	1	1	1	0	1	1	0	0	0	1	0	0	0	1	1	0	0	
S	Device Address							W	A	Command							A	P

	1	1	1	0	1	1	0	1	0	1	0	0	1	0	1	1	0	1	
S	Device Address							R	A	Data								N	P

The 4th TYPE: the host reading the 3-byte or 6-byte ADC data from the device

In this activity there are two frames that are sent separately. The first frame is identical to sending a single command, which can be one of the conversion result reading commands. In the second frame, the device will send back the ADC data (either 3 bytes or 6 bytes depending on the commands) after receiving the

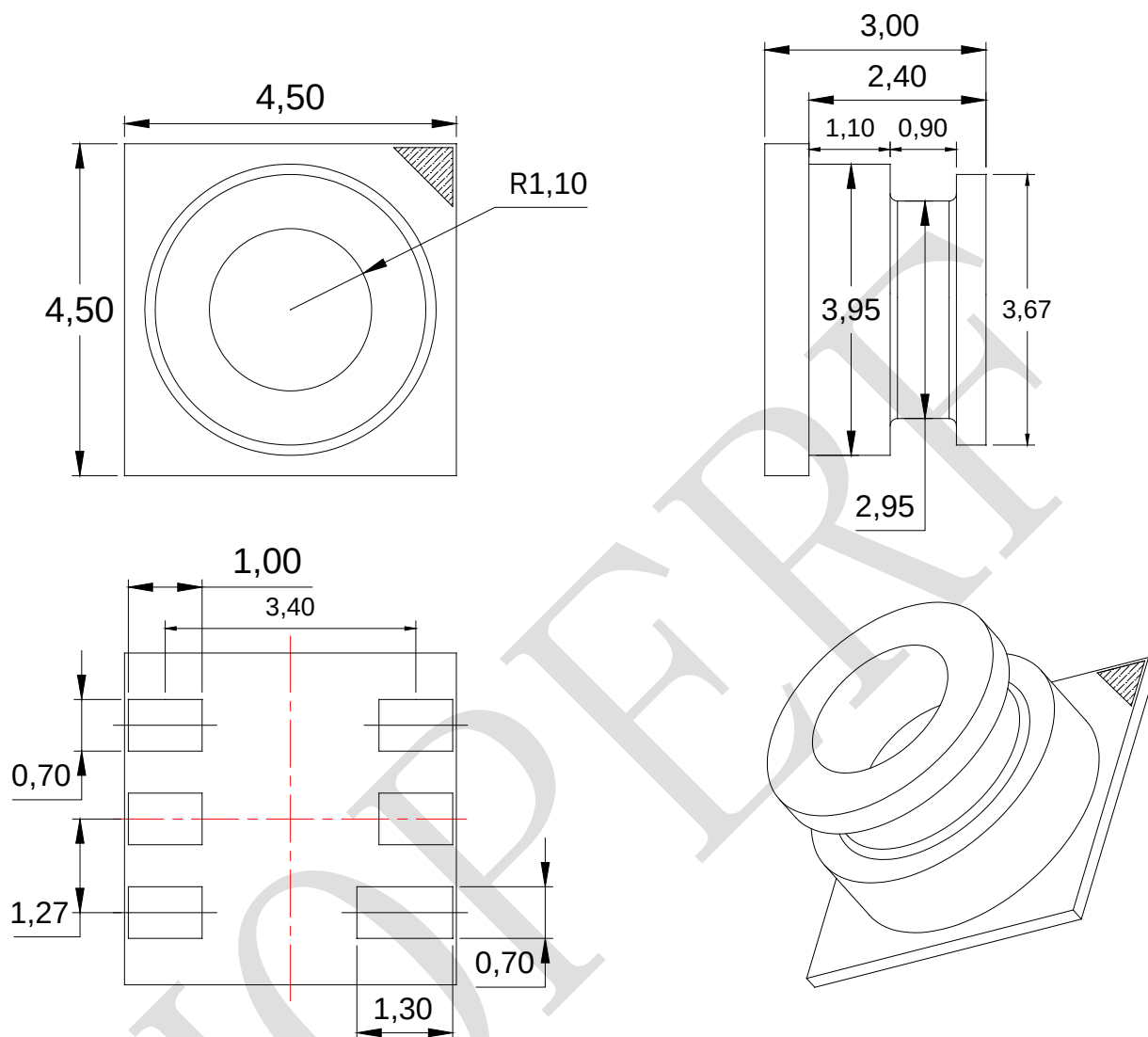
	1	1	1	0	1	1	0	0	0	0	0	0	0	0	1	1	0	0	
S	Device Address							W	A	Command								A	P

	1	1	1	0	1	1	0	1	0	0	1	0	0	0	1	1	0	0												
S	Device Address							R	A	Data Byte 6 or 3								A	Bit		0	0	1	1	0	1	0	0	1	
																		Data Byte 0								N	P			

Descriptions

☐	From Host	☐	From Chip
☐	Start Bit	☐	Stop Bit
☐	Write	☐	Read
☐	ACK	☐	NACK

7. Package information



Mechanical Dimension Unit: mm

General Tolerance: $\pm 0.1\text{mm}$

Shenzhen Hope Microelectronics Co., Ltd.

Address: 30th floor of 8th Building, C Zone, Vanke Cloud City, Xili Sub-district,
Nanshan, Shenzhen, GD, P.R. China

Tel: +86-755-82973805

Sales: sales@hoperf.com

Website: www.hoperf.com